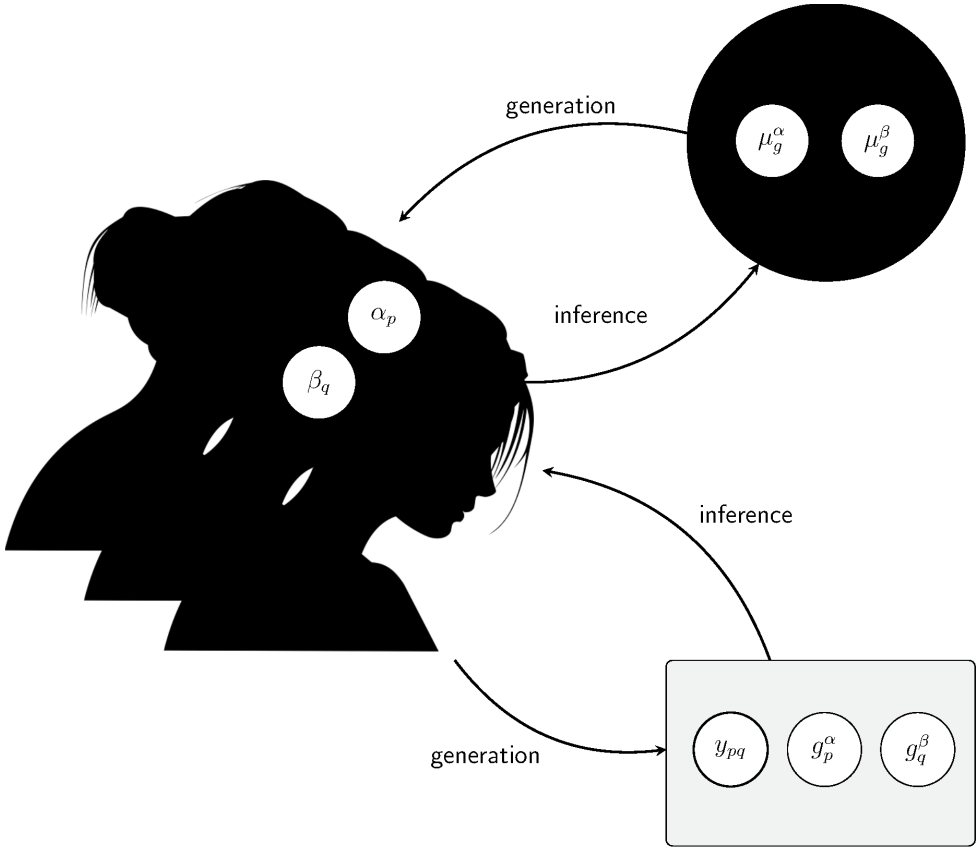
The Hierarchical Rasch Model

Michael D. Lee

Hierarchical Rasch Model Structure

- The hierarchical Rasch model has the structure
  - observed binary decisions y_{pq} for person p answering question q
  - person ability α_p and question difficulty β_q parameters
  - group assignments for both people g_p^α and questions g_q^β
  - group means (no standard deviations) for both the person and question groups



Critic Content

Critic Content Data

- The data set comes from Rotten tomatoes
- 137 regular critics and 63 top critics rate 60 movies as “fresh” or “rotten”
- Fresh ratings in blue and rotten ones in red
- Top 10 movies are rated G, next 30 are rated PG, final 20 are rated R
- Naturally occuring data with many missing entries

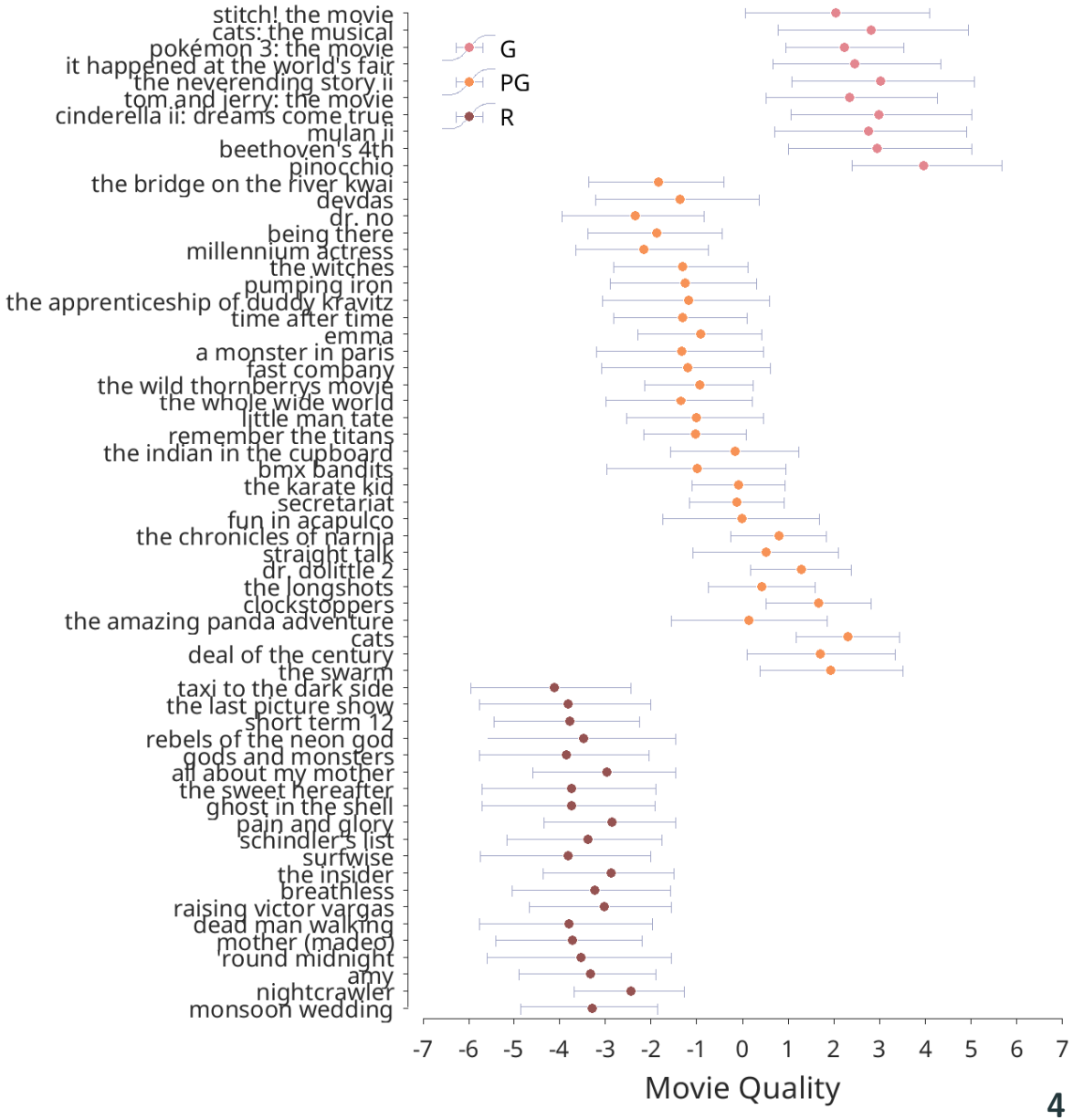


Applying the Rasch Model

- Apply the hierarchical Rasch model:
  - treat "fresh" ratings as correct
  - the distinction between regular and top critics defines two person groups
  - the three movie ratings define three question groups
- The natural interpretation of the parameters is
  - difficulty relates to the quality of a movie, with lower numbers being higher quality
  - ability relates to the standards of a critic, with higher numbers meaning they rate "fresh" more easily

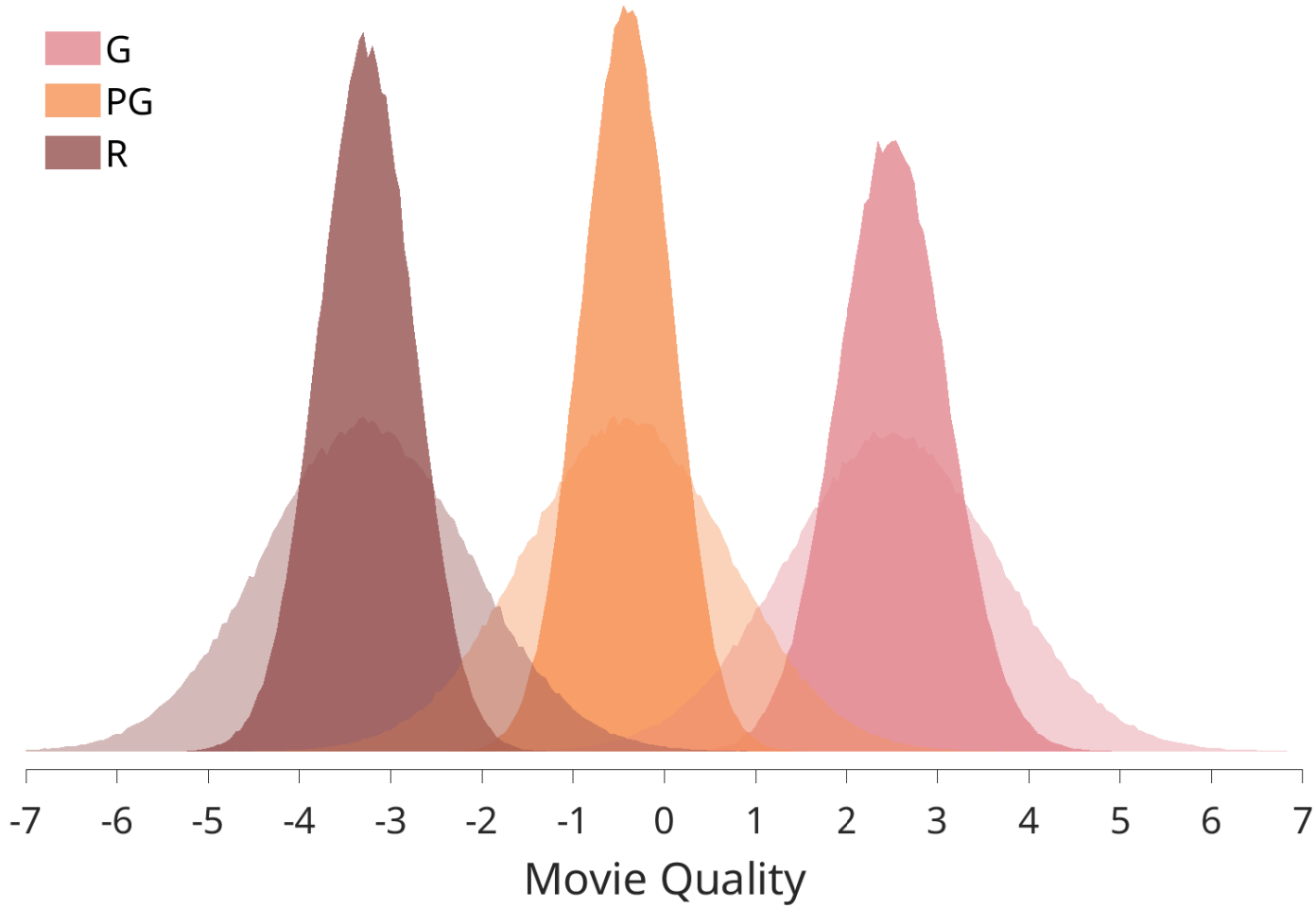
Inferred Movie Qualities

- The movies were chosen deliberately in terms of the proportion of fresh ratings in each group
- That structure is clear in the quality inferences



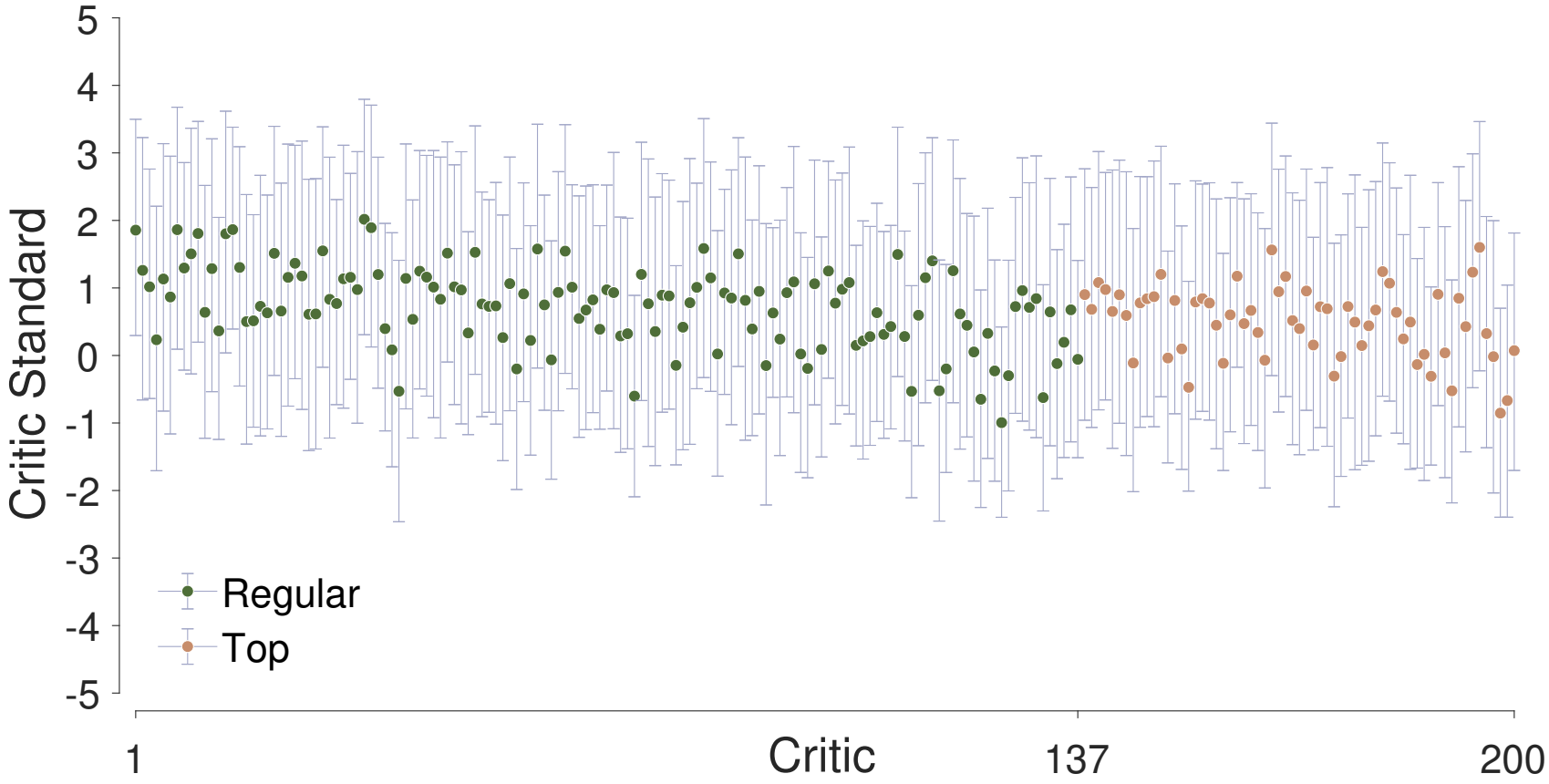
Inferred Content Rating Qualities

- The inferences about the group means and representative movies from each content class summarize the differences between them



Inferred Critic Standards

- The inferences about the standards of the individual critics show significant uncertainty, few clear individual differences, and no suggestion of group differences



Inferred Critic Type Standards

- The inferences about the group means and representative regular and top critics show no evidence of differences in their standards

